

SYLLABUS :-

1. Name of the subject Solid Mechanics CE21005 2. L-T-P 3-1-0 3. Credits 4 4. Name of the Department Civil Engineering 5. Status of the subject a Specify the Session, Semester: 2021-22 onwards b Please specify the level of the subject: B.Tech/DD c Whether the subject will be offered as core or elective: Core d The semester in which the subject will be offered: Autumn e Name of the programme in whose curricula this subject will be included: Civil Engineering B.Tech H 4 years, Civil Engineering Dual-Degree All 6. Prerequisites for the subject, if any Basic Engineering Mechanics ME11003 7. Objectives This course is concerned with the principles of solid mechanics and equations of equilibrium for deformable bodies. Different approaches for solving the equations of equilibrium are discussed. Applying these principles to the various structural elements, the resulting stresses and strains are computed. 8. Contents Minimum 36 hours and preferably in multiple of 6 hours a 39 to 42 hours for 3-0-0 or 3-1-0 b 52 to 56 hours for 4-0-0 Introduction to deformable bodies, analysis of stress, analysis of strain, constitutive relations, bending of beams, shear stresses in beams, torsion of shaft, inelastic material behaviour, energy principles, elastic stability Recommended text books and reference books 1. Engineering Mechanics of Solids by EP Popov Text book 2. Introduction to Solid Mechanics by IH Shames and JM Pitarresi Text book 3. Advanced Mechanics of Solids by LS Srinath Text book 2. Topics Number of hours Introduction to deformable bodies: Concept of stress, strain and constitutive relations 1 Analysis of stress: Definition of stress at a point, symmetry of the stress tensor/array, stress on an arbitrarily oriented plane and Cauchy's postulate, transformation of stress, stress invariants and principal stresses. Mohr's circle representation of 2D state of stress, thin-walled pressure vessel 5 Analysis of strain: Deformation of a deformable body, strain as a measure of deformation, transformation of strain, strain invariants, principal strains, Mohr's circle representation of 2D strains, strain measurement-strain rosettes. 5 Constitutive relations: Generalized Hooke's law, concept of strain energy, Elastic constants. 3 Bending of beams: Sign convention, stress resultants, simple bending, curved beams, pure bending, properties of area - principal axes and principal second moments of area, load-shear force-bending moment relationship in general, unsymmetrical bending 8 Shear stresses in beams: rectangular section, I-section, strain energy due to shear, relation between bending moment and shear force, shearing stresses in straight beam-unsymmetrical sections, the shear centre, thin-walled sections, bending of single-cell and multicell thin-walled tubes. 8 Torsion of shaft: Circular bars, strain energy in torsion, noncircular bars, warping function and Saint-Venant's stress function, bars of solid section, warping of thin-walled sections, multicell thin-walled tubes, combined bending and torsion of multicell tubes 6 Inelastic material behaviour: Ductile and brittle behaviour, yield criteria: general concepts, failure theories and safety factors. 4 The energy principles:

The principles of stationary potential energy-equilibrium, minimum potential energy-stable equilibrium, Castigliano's first theorem; The principle of stationary complementary energy, Engesser's first theorem, Castigliano's second theorem, deflections of simple structures, the Rayleigh-Betti theorem and Maxwell's theorem of reciprocity, Engesser's second theorem - theorem of least work 4 Elastic stability: Introduction to the concept of column buckling, pin-ended strut and Euler buckling load, other simple 4 - 130 -strut problems, real behaviour of struts: initial curvature, eccentric loads